SIEM Deployment & SSH Brute Force Detection

with Splunk

SOC Monitoring & Detection Project

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1. Project Overview

This project consists of building a small Security Operations Center (SOC) laboratory using Splunk Enterprise as a Security Information and Event Management (SIEM) platform.

The main objective is to collect authentication logs from a Kali Linux machine, forward them to a centralized Splunk server, analyze the logs, detect repeated failed SSH login attempts, generate a security alert, and visualize the activity through a SOC dashboard.

The project demonstrates an end-to-end security monitoring workflow, from log generation and collection to detection, alerting, and visualization.

2. Project Objectives

The main objectives of the project are:

- Deploy Splunk Enterprise as a SIEM platform.
- Configure a Splunk Universal Forwarder.
- Collect Linux authentication logs.
- Forward authentication events to the SIEM.
- Analyze SSH authentication activity using SPL.
- Detect repeated failed SSH login attempts.
- Configure an automated security alert.
- Build a monitoring dashboard for SOC analysis.

3. Lab Architecture

The laboratory consists of two main virtual machines.

- **Kali Linux:** used as the log source and for controlled SSH authentication testing.
- **Ubuntu Server:** used as the centralized Splunk Enterprise SIEM server.

The communication between the machines is performed through an isolated private virtual network.

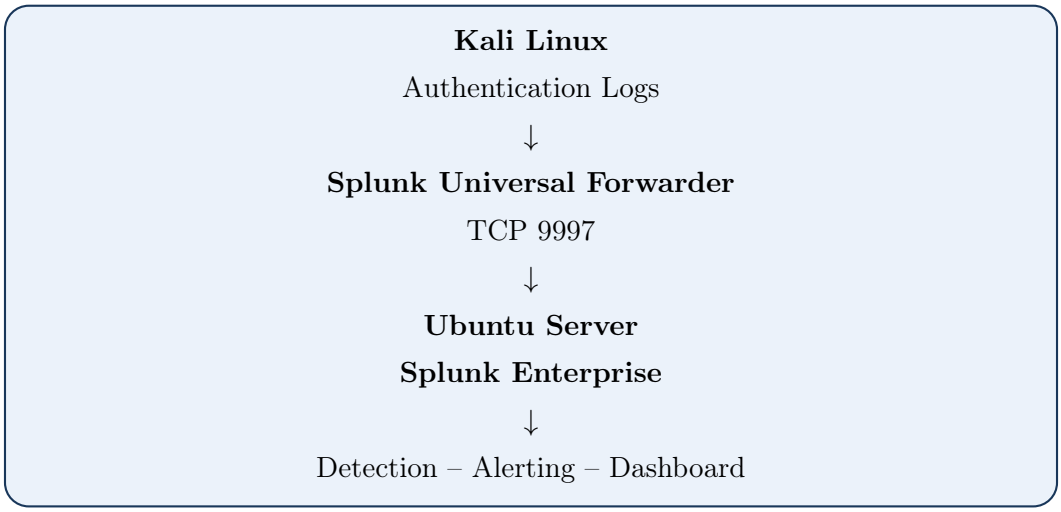


Figure 1: SOC laboratory architecture

4. Technologies Used

Technology	Purpose
Kali Linux	Log source and security testing
Ubuntu Server	SIEM server
Splunk Enterprise 10.4.3	SIEM platform
Splunk Universal Forwarder 10.4.3	Log forwarding
SSH	Authentication monitoring
SPL	Log analysis and detection
VirtualBox	Virtualization

Table 1: Technologies used in the project

5. Log Collection and Forwarding

The authentication log monitored on Kali Linux is:

```
/var/log/auth.log
```

The Splunk Universal Forwarder was configured to monitor the authentication log and forward generated security events to Splunk Enterprise.

The complete log collection workflow is:

SSH Activity

↓

/var/log/auth.log

↓

Splunk Universal Forwarder

↓

TCP 9997

↓

Splunk Enterprise

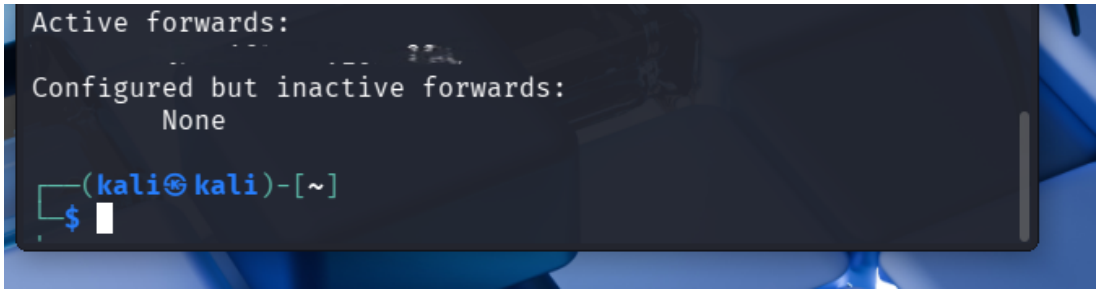


Figure 2: Universal Forwarder showing an active forward

6. Log Analysis with SPL

Splunk Search Processing Language (SPL) was used to analyze authentication events collected from Kali Linux.

The first search was used to identify failed SSH authentication attempts:

```
"Failed password"
```

This query searches the collected authentication events for failed password attempts.

The collected events allowed the analysis of:

- Failed authentication attempts.
- Source IP addresses.
- Target usernames.
- Authentication activity over time.

7. SSH Brute Force Detection

A detection rule was created to identify repeated failed SSH authentication attempts.

The detection logic used was:

```
"Failed password"
| rex "Failed password for (?:invalid user )?(?<user>\S+) from (?<src_ip>\S+)"
| bin _time span=5m
| stats count by _time src_ip user
| where count >= 3
| sort - count
```

7.1. Detection Logic

The detection performs the following operations:

1. Searches authentication events.
2. Filters failed SSH password attempts.
3. Extracts the username and source IP address.

4. Groups events into five-minute time windows.
5. Counts authentication failures.
6. Identifies cases with three or more failed attempts.

The detection threshold used in this laboratory is:

3 or more failed SSH attempts within 5 minutes

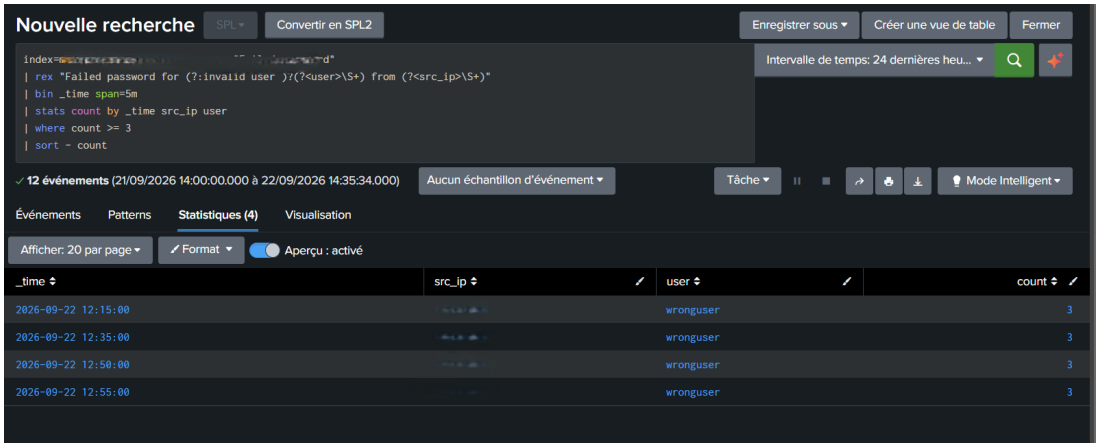


Figure 3: SSH brute-force detection in Splunk

8. Security Alert

A Splunk alert named:

SSH Brute Force Detection

was configured using the detection logic.

The alert was scheduled to execute every five minutes and analyze the most recent five-minute period.

8.1. Alert Configuration

Parameter	Configuration
Alert Name	SSH Brute Force Detection
Schedule	Every 5 minutes
Search Window	Last 5 minutes
Trigger Condition	Number of results > 0
Trigger Frequency	Once
Throttle	300 seconds
Severity	Medium

Table 2: SSH brute-force alert configuration

The alert was successfully triggered during testing.

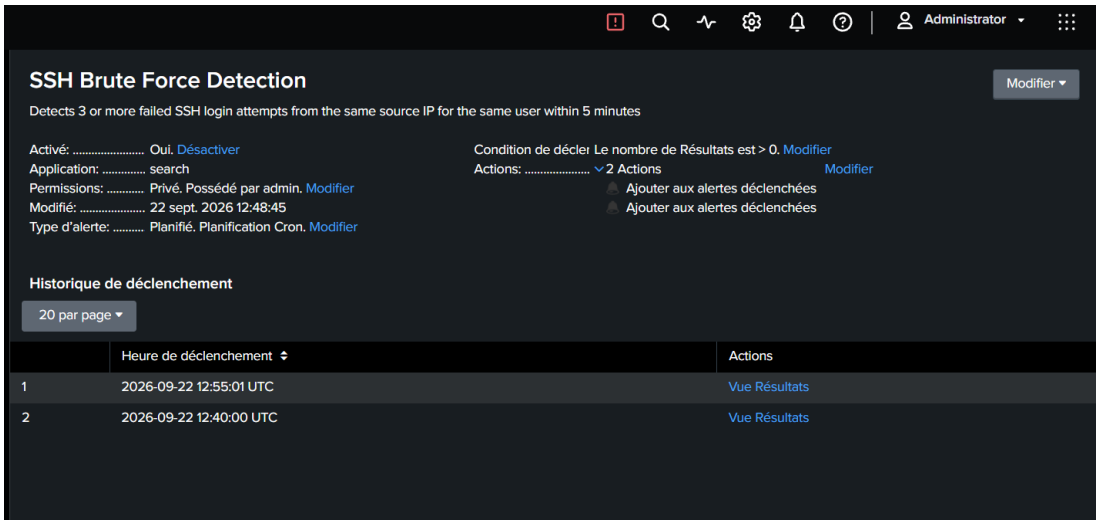


Figure 4: Triggered SSH brute-force security alert

9. Testing and Results

A controlled SSH authentication test was performed in the laboratory environment.

The following command was used:

```
ssh wronguser@127.0.0.1
```

Three incorrect authentication attempts were generated.

Splunk successfully identified the activity and produced the following detection result:

Parameter	Detected Value
Source	Localhost
Target User	wronguser
Failed Attempts	3
Detection Window	5 minutes
Detection Status	Triggered

Table 3: SSH brute-force test result

The test confirmed that the complete monitoring pipeline was operational:

SSH Activity

→

Log Collection

→

Splunk

→

Detection

→

Alert

10. SOC Dashboard

A SOC monitoring dashboard was created in Splunk to provide a visual overview of authentication activity.

The dashboard contains four monitoring panels:

1. **Failed SSH Login Timeline**
2. **Total Failed SSH Attempts**
3. **Top Source IPs**
4. **SSH Brute Force Detections**

The dashboard provides a centralized view of authentication activity and helps identify repeated failed login attempts.

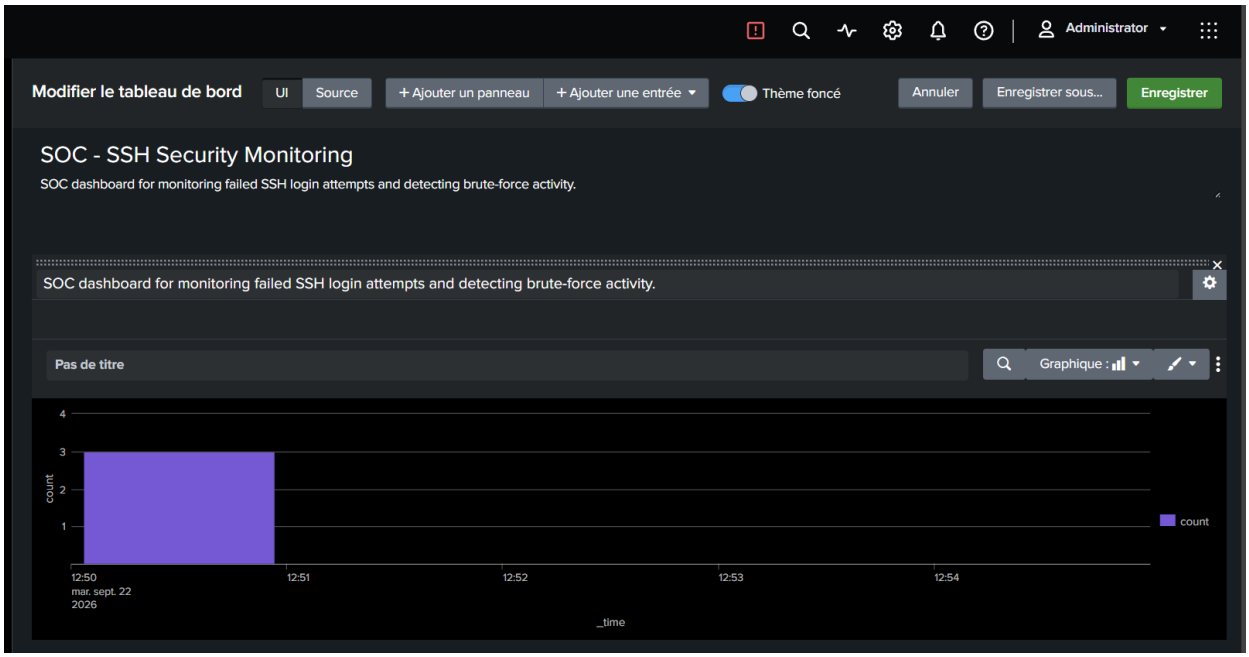


Figure 5: SOC monitoring dashboard

11. Project Results

The project successfully achieved the main objectives:

- Authentication logs were collected from Kali Linux.
- Logs were forwarded to Splunk Enterprise.
- SPL queries were used to analyze authentication events.
- Repeated failed SSH attempts were detected.
- A brute-force detection rule was implemented.
- A scheduled security alert was configured.
- The alert was successfully triggered during testing.
- A SOC monitoring dashboard was created.

12. Skills Demonstrated

This project provided practical experience in:

- SIEM deployment and configuration.
- Splunk Enterprise administration.
- Splunk Universal Forwarder configuration.
- Linux authentication log analysis.
- SPL query development.
- SSH security monitoring.
- Brute-force detection.
- Security alert configuration.
- SOC dashboard creation.
- Basic security event investigation.